

ScienceQtech Employees Performance Mapping SQL Project Report

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1. Executive Summary

ScienceQtech Employees Performance Mapping – SQL Project is a MySQL-based data analysis project designed to analyze employee performance, compensation, experience, project allocation, and Data Science team membership at ScienceQtech, a data science-focused startup. The project uses three related datasets: emp_record_table, proj_table, and data_science_team. The original project brief describes the objective as analyzing employee details, performance, and projects undertaken for the HR analysis function.

The project includes 16 SQL-based business questions covering employee information, performance ratings, leadership roles, salary analysis, experience ranking, views, indexing and execution plans, salary distribution, Data Science team analysis, project duration, employee-project mapping, and project-domain allocation.

An Entity Relationship Diagram (ERD) was also created to represent the relationships between the employee, project, and Data Science team tables and to provide a visual representation of the database structure.

The analysis demonstrates the use of SQL techniques including SELECT statements, filtering, aggregation, GROUP BY, window functions, views, indexes, execution plans, JOINS, CASE statements, CTEs, subqueries, and date functions.

2. Question-wise Output / Results

2.1. Create a database named employee, then import data_science_team.csv, proj_table.csv, and emp_record_table.csv into the employee database from the given resources.

Result:

The employee database was created and the three project datasets—emp_record_table, proj_table, and data_science_team—were imported into the database.

2.2 Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, and DEPARTMENT from the employee record table, and make a list of employees and details of their department.

Screenshot:



The screenshot shows a MySQL query result with the following columns: EMP_ID, FIRST_NAME, LAST_NAME, GENDER, and DEPT. The data is as follows:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT
E001	Arthur	Black	M	ALL
E005	Eric	Hoffman	M	FINANCE
E010	William	Butler	M	AUTOMOTIVE
E052	Dianna	Wilson	F	HEALTHCARE
E057	Dorothy	Wilson	F	HEALTHCARE
E083	Patrick	Voltz	M	HEALTHCARE
E103	Emily	Grove	F	FINANCE
E204	Karene	Nowak	F	AUTOMOTIVE
E245	Nian	Zhen	M	RETAIL
E260	Roy	Collins	M	RETAIL
E403	Steve	Hoffman	M	FINANCE
E428	Pete	Allen	M	AUTOMOTIVE
E478	David	Smith	M	RETAIL

Result:

The output displays employee IDs, names, gender, and departments for the employees in the employee record table.

2.3 Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPARTMENT, and EMP_RATING if the EMP_RATING is:

- less than two
- between two and four
- greater than four

Screenshot:

	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT	EMP_RATING
▶	E001	Arthur	Black	M	ALL	5
	E005	Eric	Hoffman	M	FINANCE	3
	E010	William	Butler	M	AUTOMOTIVE	2
	E052	Dianna	Wilson	F	HEALTHCARE	5
	E057	Dorothy	Wilson	F	HEALTHCARE	1
	E083	Patrick	Voltz	M	HEALTHCARE	5
	E103	Emily	Grove	F	FINANCE	4
	E204	Karene	Nowak	F	AUTOMOTIVE	5
	E245	Nian	Zhen	M	RETAIL	2
	E260	Roy	Collins	M	RETAIL	3
	E403	Steve	Hoffman	M	FINANCE	3
	E428	Pete	Allen	M	AUTOMOTIVE	4
	E478	David	Smith	M	RETAIL	4

Result:

The output displays employee details along with their performance ratings, covering ratings below 2, between 2 and 4, and above 4.

2.4 Write a SQL query to retrieve the employee ID, first name, role, and department of employees who hold leadership positions (Manager, President, or CEO).

Screenshot:

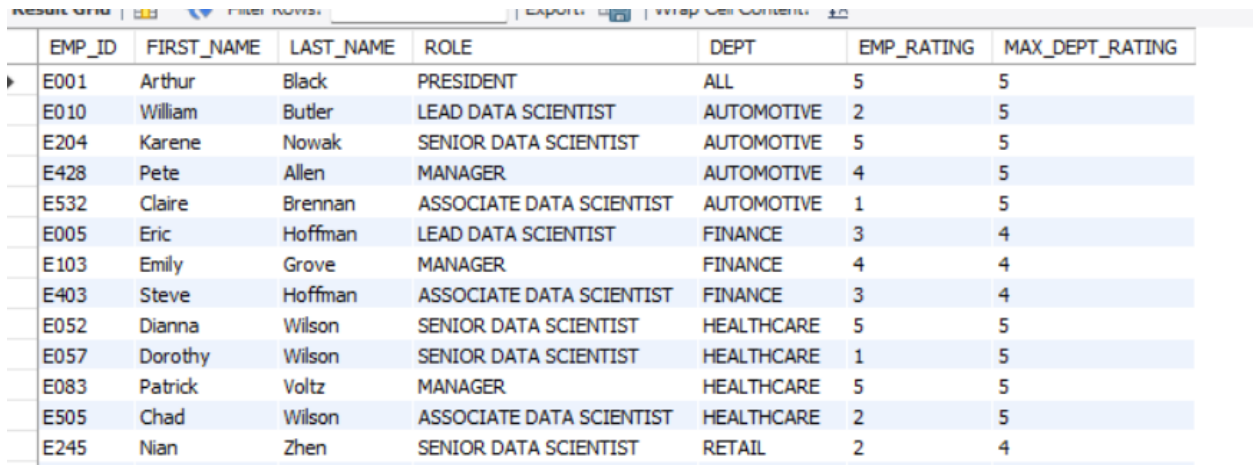
Result Grid				
	EMP_ID	FIRST_NAME	ROLE	DEPT
▶	E001	Arthur	PRESIDENT	ALL
	E083	Patrick	MANAGER	HEALTHCARE
	E103	Emily	MANAGER	FINANCE
	E428	Pete	MANAGER	AUTOMOTIVE
	E583	Janet	MANAGER	RETAIL
	E612	Tracy	MANAGER	RETAIL
•	NULL	NULL	NULL	NULL

Result:

The output identifies employees holding leadership roles, including the President and Managers, along with their employee IDs, names, and departments.

2.5 Write a query to list employee details, including EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPARTMENT, and EMP_RATING, grouped by department. Also include the respective employee rating along with the max emp rating for the department.

Screenshot:



The screenshot shows a SQL query result grid with columns: EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, EMP_RATING, and MAX_DEPT_RATING. The data is grouped by department (DEPT). The departments listed are ALL, AUTOMOTIVE, FINANCE, HEALTHCARE, and RETAIL. The MAX_DEPT_RATING column shows the maximum rating for each department.

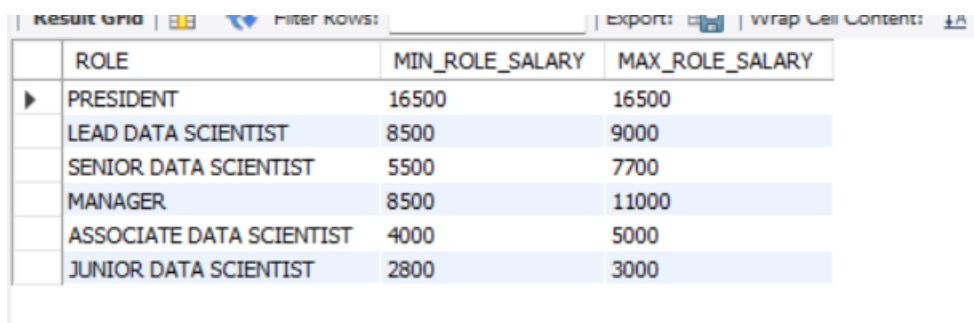
EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	EMP_RATING	MAX_DEPT_RATING
E001	Arthur	Black	PRESIDENT	ALL	5	5
E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	2	5
E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	5	5
E428	Pete	Allen	MANAGER	AUTOMOTIVE	4	5
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	1	5
E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	3	4
E103	Emily	Grove	MANAGER	FINANCE	4	4
E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	FINANCE	3	4
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	5	5
E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	1	5
E083	Patrick	Voltz	MANAGER	HEALTHCARE	5	5
E505	Chad	Wilson	ASSOCIATE DATA SCIENTIST	HEALTHCARE	2	5
E245	Nian	Zhen	SENIOR DATA SCIENTIST	RETAIL	2	4

Result:

The output displays each employee's performance rating alongside the maximum employee rating recorded within their respective department.

2.6 Write a query to find the minimum and maximum salary for employees grouped by their role, using data from the employee record table.

Screenshot:



The screenshot shows a SQL query result grid with columns: ROLE, MIN_ROLE_SALARY, and MAX_ROLE_SALARY. The data is grouped by role (ROLE). The roles listed are PRESIDENT, LEAD DATA SCIENTIST, SENIOR DATA SCIENTIST, MANAGER, ASSOCIATE DATA SCIENTIST, and JUNIOR DATA SCIENTIST. The MIN_ROLE_SALARY and MAX_ROLE_SALARY columns show the minimum and maximum salary for each role.

ROLE	MIN_ROLE_SALARY	MAX_ROLE_SALARY
PRESIDENT	16500	16500
LEAD DATA SCIENTIST	8500	9000
SENIOR DATA SCIENTIST	5500	7700
MANAGER	8500	11000
ASSOCIATE DATA SCIENTIST	4000	5000
JUNIOR DATA SCIENTIST	2800	3000

Result: The output shows the minimum and maximum salary for each role, ranging from ₹2,800–₹3,000 for Junior Data Scientists to ₹16,500 for the President.

2.7 Write a query to assign a rank to each employee based on their years of experience, using data from the employee record table.

Screenshot:

	EMP_ID	FIRST_NAME	LAST_NAME	EXP_RANK
▶	E001	Arthur	Black	1
	E083	Patrick	Voltz	2
	E103	Emily	Grove	3
	E428	Pete	Allen	3
	E583	Janet	Hale	3
	E612	Tracy	Norris	4
	E010	William	Butler	5
	E005	Eric	Hoffman	6
	E057	Dorothy	Wilson	7
	E204	Karene	Nowak	8
	E260	Roy	Collins	9
	E052	Dianna	Wilson	10
	E245	Nian	Zhen	10

Result:

The output ranks employees according to their years of experience, with the highest experience receiving rank 1. Employees with equal experience share the same rank.

2.8 Write a query to create a view that displays employees in various countries whose salary is more than six thousand, using data from the employee record table.

Screenshot:

	EMP_ID	FIRST_NAME	LAST_NAME	SALARY	COUNTRY
▶	E001	Arthur	Black	16500	USA
	E005	Eric	Hoffman	8500	USA
	E010	William	Butler	9000	FRANCE
	E057	Dorothy	Wilson	7700	USA
	E083	Patrick	Voltz	9500	USA
	E103	Emily	Grove	10500	CANADA
	E204	Karene	Nowak	7500	GERMANY
	E245	Nian	Zhen	6500	CHINA
	E260	Roy	Collins	7000	INDIA
	E428	Pete	Allen	11000	GERMANY
	E583	Janet	Hale	10000	COLOMBIA
	E612	Tracy	Norris	8500	INDIA

Result:

The output displays employees earning more than 6,000, along with their employee IDs, names, salaries, and countries.

2.9 Create an index to improve the cost and performance of the query to find the employee whose FIRST_NAME is Eric in the employee table, after checking the execution plan.

Screenshots:

Result Grid												
	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
▶	1	SIMPLE	emp_record_table	NULL	ALL	NULL	NULL	NULL	NULL	19	10.00	Using where

Result Grid												
	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
▶	1	SIMPLE	emp_record_table	NULL	ref	idx_first_name	idx_first_name	203	const	1	100.00	Using where

Result:

Before indexing, the query performed a full table scan across 19 rows. After creating idx_first_name, the execution plan used an index lookup and reduced the estimated rows examined to 1.

2.10 Write a query to calculate the average salary distribution based on the continent and country. Take data from the employee record table.

Screenshot:

Result Grid			
	CONTINENT	COUNTRY	AVG_SALARY
▶	ASIA	CHINA	6500.0000
	ASIA	INDIA	6166.6667
	EUROPE	FRANCE	9000.0000
	EUROPE	GERMANY	7600.0000
	NORTH AMERICA	CANADA	7000.0000
	NORTH AMERICA	USA	9440.0000
	SOUTH AMERICA	COLOMBIA	5600.0000

Result:

The output displays average salary for each country grouped within its respective continent, with the USA showing an average salary of 9,440 among the listed countries.

2.11 Which departments have the most experienced data-science employees on average?

Screenshot:

Result Grid			Filter Rows:		Export:		Wr
	DEPT	AVG_EXP	MIN_EXP	MAX_EXP	TOTAL_EMP		
	AUTOMOTIVE	7.67	3	12	3		
	FINANCE	7.50	4	11	2		
	HEALTHCARE	6.67	5	9	3		
	RETAIL	3.80	1	7	5		

Result:

The output shows average, minimum, and maximum experience along with employee counts for each department. Automotive has the highest average experience at 7.67 years.

2.12 Write a query to classify data science team employees into Junior, Middle-Level, and Senior categories based on their years of experience, while displaying their employee details.

Screenshot:

Result Grid				Filter Rows		Export		Wrap Cell Content		
	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	EXP_LEVEL
▶	E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	Senior
	E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	Senior
	E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	Senior
	E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	Senior
	E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA	Middle
	E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	Middle
	E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	Middle
	E505	Chad	Wilson	M	ASSOCIATE DATA SCIENTIST	HEALTHCARE	5	CANADA	NORTH AMERICA	Middle
	E403	Steve	Hoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE	4	USA	NORTH AMERICA	Middle
	E478	David	Smith	M	ASSOCIATE DATA SCIENTIST	RETAIL	3	COLOMBIA	SOUTH AMERICA	Junior
	E532	Claire	Brennan	F	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	3	GERMANY	EUROPE	Junior
	E620	Katrina	Allen	F	JUNIOR DATA SCIENTIST	RETAIL	2	INDIA	ASIA	Junior
	E640	Jenifer	Jhones	F	JUNIOR DATA SCIENTIST	RETAIL	1	COLOMBIA	SOUTH AMERICA	Junior

Result:

The Data Science team employees are classified into **Junior, Middle, and Senior** categories based on their years of experience, with their employee details displayed alongside the classification.

2.13 Using a CTE, calculate the duration of each project in days. Then, using a subquery, calculate the average project duration and display the projects whose duration is greater than the overall average.

Screenshot:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

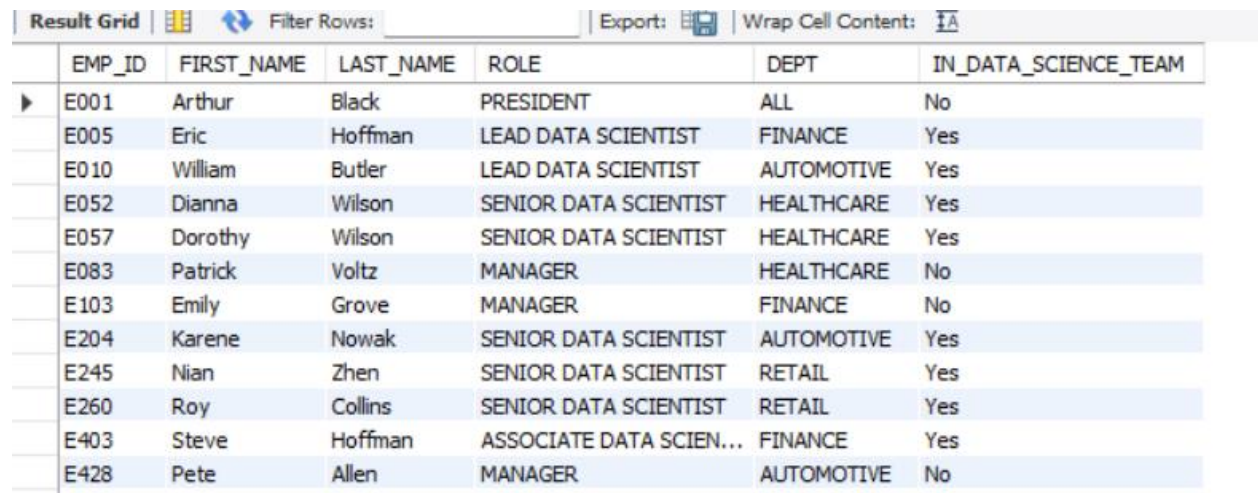
	PROJECT_ID	PROJ_NAME	DOMAIN	START_DATE	DEV_QTR	STATUS	CLOSURE_DATE	DURATION_DAYS
	P109	Market Basket Analysis	RETAIL	2021-04-12	Q1	DELAYED	2021-06-30	79
	P406	Customer Sentiment Analysis	RETAIL	2021-07-09	Q2	WIP	2021-09-24	77

Result:

The output identifies projects whose duration is greater than the overall average project duration. Market Basket Analysis has a duration of 79 days, while Customer Sentiment Analysis has a duration of 77 days.

2.14 Using employee records and Data Science team records, display all employees and indicate whether each employee belongs to the Data Science team.

Screenshot:



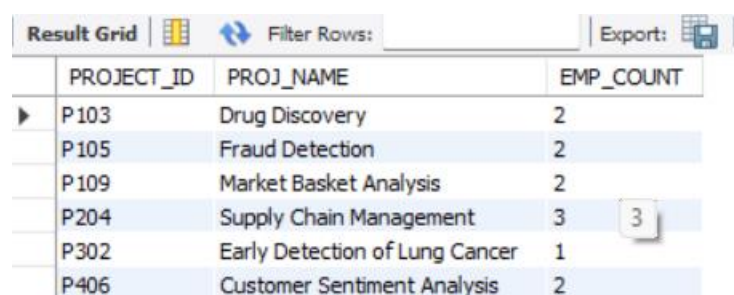
EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	IN_DATA_SCIENCE_TEAM
E001	Arthur	Black	PRESIDENT	ALL	No
E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	Yes
E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	Yes
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	Yes
E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	Yes
E083	Patrick	Voltz	MANAGER	HEALTHCARE	No
E103	Emily	Grove	MANAGER	FINANCE	No
E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	Yes
E245	Nian	Zhen	SENIOR DATA SCIENTIST	RETAIL	Yes
E260	Roy	Collins	SENIOR DATA SCIENTIST	RETAIL	Yes
E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	FINANCE	Yes
E428	Pete	Allen	MANAGER	AUTOMOTIVE	No

Result:

The output lists all employees and indicates whether each employee belongs to the Data Science team using a Yes/No classification.

2.15 Using emp_record_table and proj_table, calculate the number of employees assigned to each project. Display the project ID, project name, and employee count.

Screenshot:



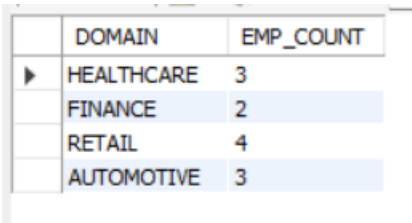
PROJECT_ID	PROJ_NAME	EMP_COUNT
P103	Drug Discovery	2
P105	Fraud Detection	2
P109	Market Basket Analysis	2
P204	Supply Chain Management	3
P302	Early Detection of Lung Cancer	1
P406	Customer Sentiment Analysis	2

Result:

The output shows the number of employees assigned to each project. Supply Chain Management has the highest employee count at 3, while Early Detection of Lung Cancer has 1 assigned employee.

2.16 Using emp_record_table and proj_table, calculate the number of employees working in each project domain. Display the project domain and employee count.

Screenshot:



A screenshot of a database query result displayed in a window. The result is a table with two columns: 'DOMAIN' and 'EMP_COUNT'. The rows are: HEALTHCARE (3), FINANCE (2), RETAIL (4), and AUTOMOTIVE (3). The rows are alternatingly highlighted with light blue and white backgrounds. A small black triangle icon is visible to the left of the first row.

	DOMAIN	EMP_COUNT
▶	HEALTHCARE	3
	FINANCE	2
	RETAIL	4
	AUTOMOTIVE	3

Result:

The output shows employee counts by project domain, with Retail having 4 employees, followed by Healthcare and Automotive with 3 each, and Finance with 2.